

Bitcoin Node types

- 1) Wallet
- 2) light (less)
- 3) Miner (half)
- 4) Full (complete)

Lecture-1

Lecture-2

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **A cryptographic hash function must always produce a fixed-size output.**
 - True
 2. **The output of a cryptographic hash function can be of variable length depending on the input.**
 - **False**
Corrected version: The output of a cryptographic hash function is always of fixed size.
 3. **Collision resistance is a desirable property for a cryptographic hash function, meaning it is impossible to find two different inputs that hash to the same value.**
 - True
 4. **A cryptographic hash function must be slow to compute to ensure security.**
 - **False**
Corrected version: A cryptographic hash function should be fast to compute, as it must be efficiently computable.
 5. **The "pre-image resistance" property of a hash function makes it difficult to reverse the function and find the original input.**
 - True
-

2. Fill in the blanks: Complete the sentences.

6. **A cryptographic hash function must be able to accept any string as input and produce a _____ output.**
 - **Answer:** fixed-size.
 7. **The pre-image resistance property makes it _____ to find an input that hashes to a specific value.**
 - **Answer:** computationally infeasible.
 8. **The term "puzzle-friendliness" in a hash function means that it is hard to find an input that results in a hash within a _____.**
 - **Answer:** target set.
 9. **Collision resistance in a hash function means that it is infeasible to find two distinct _____ that produce the same hash.**
 - **Answer:** inputs.
 10. **In cryptographic hash functions, "min-entropy" is a measure of how _____ the outcome is.**
 - **Answer:** predictable.
-

3. Multiple Choice: Choose the correct answer.

11. **What is the primary purpose of a cryptographic hash function?**
 - a) To ensure that the hash size matches the input size
 - b) To produce a fixed-size output from a variable input
 - c) To provide a reversible function
 - **Answer:** b) To produce a fixed-size output from a variable input
12. **Which of the following is a desirable property of a cryptographic hash function?**
 - a) Efficiency
 - b) Reversibility
 - c) Collision resistance
 - **Answer:** c) Collision resistance
13. **What is "pre-image resistance" in the context of cryptographic hash functions?**
 - a) It ensures that finding a hash value is computationally infeasible
 - b) It makes it infeasible to find an input that produces a specific hash
 - c) It guarantees that two different inputs produce different hashes
 - **Answer:** b) It makes it infeasible to find an input that produces a specific hash
14. **Which of the following is the correct statement about SHA-256?**
 - a) It is a symmetric encryption algorithm
 - b) It is a collision-resistant hash function
 - c) It is a public-key encryption algorithm
 - **Answer:** b) It is a collision-resistant hash function
15. **Which property of a cryptographic hash function makes it infeasible to find two different inputs that hash to the same output?**
 - a) Pre-image resistance
 - b) Collision resistance

- c) Puzzle-friendliness
 - **Answer:** b) Collision resistance
16. **What is the expected number of attempts needed to find a collision in a hash function with a 256-bit output?**
- a) 2^{128}
 - b) 2^{256}
 - c) 2^{64}
 - **Answer:** a) 2^{128}
-

4. Short Answer Questions

17. **What is the birthday paradox, and how does it relate to cryptographic hash functions?**
- **Answer:** The birthday paradox states that the probability of two people having the same birthday exceeds 50% with just 23 people. Similarly, for cryptographic hash functions, it shows that the probability of finding two different inputs that produce the same hash is much higher than expected, which leads to the concept of collision resistance.
18. **Explain the concept of "min-entropy" in relation to cryptographic hash functions.**
- **Answer:** Min-entropy refers to the unpredictability of a random variable. In the context of cryptographic hash functions, high min-entropy implies that the output of the hash function is highly unpredictable and resistant to attacks.
19. **What is the Merkle-Damgård construction used in hash functions like SHA-256?**
- **Answer:** The Merkle-Damgård construction is a method used to build hash functions where the message is processed in fixed-size blocks. The message is first padded to a multiple of the block size, and then each block is hashed iteratively, producing a final hash value.
20. **How does a collision-resistant hash function benefit blockchain technology?**
- **Answer:** Collision resistance ensures that no two different pieces of data will have the same hash. This is crucial for blockchain as it prevents the possibility of two conflicting transactions being accepted as valid.
21. **What is a "message digest" in cryptographic hashing?**
- **Answer:** A message digest is the fixed-size output produced by a hash function after processing an input. It acts as a digital fingerprint for the input data, ensuring integrity and authenticity.
22. **What does it mean for a cryptographic hash function to be "puzzle-friendly"?**
- **Answer:** A puzzle-friendly hash function ensures that finding a solution to a puzzle (by finding a specific hash value) is computationally difficult and requires substantial work, often involving a brute-force search.
23. **How is SHA-256 related to Merkle-Damgård construction?**
- **Answer:** SHA-256 is based on the Merkle-Damgård construction, where it processes the input in 512-bit blocks, iterates through the blocks with the use of a compression function, and outputs a 256-bit hash.

24. What is the role of the "key" in a commitment scheme using cryptographic hash functions?

- **Answer:** The key in a commitment scheme is a random value used in conjunction with the message to create a hash (commitment) that hides the message. The commitment can later be revealed, proving the message without allowing for manipulation.

25. Why is SHA-256 considered collision-resistant?

- **Answer:** SHA-256 is considered collision-resistant because it is computationally infeasible to find two distinct inputs that hash to the same output. This makes it highly secure for applications like Bitcoin.

26. How does the difficulty adjustment in Bitcoin mining relate to hash functions?

- **Answer:** The difficulty adjustment in Bitcoin mining ensures that new blocks are mined approximately every 10 minutes. This is done by adjusting the difficulty of the Proof of Work puzzle, which relies on finding a hash below a certain target, making the mining process more challenging as hashing power increases.

27. What are some of the applications of cryptographic hash functions in blockchain systems?

- **Answer:** Cryptographic hash functions are used for transaction verification, creating digital signatures, maintaining the integrity of blockchain data, ensuring collision resistance, and generating unique transaction IDs.

28. What happens if a collision occurs in a cryptographic hash function?

- **Answer:** If a collision occurs, it means two different inputs produce the same hash. This breaks the integrity of the hash function and can lead to security vulnerabilities, such as the possibility of altering data while keeping the hash value the same.

29. Why is pre-image resistance important in cryptographic hash functions?

- **Answer:** Pre-image resistance ensures that it is computationally infeasible to reverse a hash function and retrieve the original input from the output, providing security against reverse-engineering and protecting the privacy of the input data.

30. What role does cryptographic hashing play in ensuring the integrity of transactions in blockchain?

- **Answer:** Cryptographic hashing ensures transaction integrity by generating a unique hash for each transaction. Any alteration in the transaction data would result in a completely different hash, making tampering detectable and ensuring the immutability of blockchain records.

Lecture-3

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **A hash pointer points to the location of data and contains a cryptographic hash of the data.**
 - True
 2. **A cryptographic hash function maps a large input space to a larger output space.**
 - False
Corrected version: A cryptographic hash function maps a large input space to a smaller, fixed-size output space.
 3. **A hash pointer can only be used to verify data, not to retrieve it.**
 - False
Corrected version: A hash pointer is used to retrieve the data and verify its integrity.
 4. **A Merkle Tree uses hash pointers and is an efficient way to verify large data structures.**
 - True
 5. **A Bloom filter always provides the correct result when checking for membership.**
 - False
Corrected version: A Bloom filter is probabilistic and may return false positives.
-

2. Fill in the blanks: Complete the sentences.

6. **A cryptographic hash function must produce a _____ output.**
 - Answer: fixed-size.
 7. **In a Merkle Tree, the leaf nodes contain _____ data, while the internal nodes contain _____.**
 - Answer: data, hashes.
 8. **A Bloom filter is used to test if an element is a member of a _____.**
 - Answer: set.
 9. **The security of a Bloom filter is based on the use of _____ hash functions.**
 - Answer: multiple.
 10. **In the context of public key cryptography, the public key is used for _____, and the private key is used for _____.**
 - Answer: encryption, decryption.
-

3. Multiple Choice: Choose the correct answer.

11. **Which of the following is a property of a cryptographic hash function?**

- a) It maps a large input space to a fixed-size output
 - b) It is reversible
 - c) It produces variable-length outputs
 - **Answer:** a) It maps a large input space to a fixed-size output
12. **What is the main advantage of using a Merkle Tree in blockchain systems?**
- a) It allows faster transaction verification
 - b) It ensures privacy for the transaction data
 - c) It increases the size of the blockchain
 - **Answer:** a) It allows faster transaction verification
13. **Which of the following is the key benefit of a Bloom filter?**
- a) It uses a small amount of memory
 - b) It eliminates all false positives
 - c) It is deterministic
 - **Answer:** a) It uses a small amount of memory
14. **What is the main limitation of a Bloom filter?**
- a) It cannot be used to test set membership
 - b) It always returns the correct result
 - c) It may produce false positives
 - **Answer:** c) It may produce false positives
15. **What type of data structure is a Merkle Tree?**
- a) Linear
 - b) Graph-based
 - c) Tree-based
 - **Answer:** c) Tree-based
-

4. Short Answer Questions

16. **What is the difference between a cryptographic hash function and a traditional hash function?**
- **Answer:** A cryptographic hash function is designed to have security properties such as collision resistance, pre-image resistance, and hiding, while traditional hash functions may not provide such security guarantees.
17. **How does a hash pointer help ensure the integrity of data?**
- **Answer:** A hash pointer stores the location of data along with a cryptographic hash of the data, allowing anyone to verify that the data hasn't been tampered with by comparing the hash to the expected value.
18. **What is the role of a Merkle Tree in blockchain technology?**
- **Answer:** A Merkle Tree is used to efficiently and securely verify that a certain data block is contained in a larger set (e.g., in the blockchain) by using hash pointers and reducing the amount of data required for verification.
19. **Why are false positives a concern when using Bloom filters?**

- **Answer:** False positives can occur in Bloom filters, meaning the filter might indicate that an element is in the set when it is not, which can lead to incorrect decisions. However, false negatives do not occur.
20. **How does a Bloom filter work to test set membership?**
- **Answer:** A Bloom filter uses multiple hash functions to map elements to positions in a bit array. When testing membership, it checks the corresponding positions; if all are set to 1, the element is probably in the set. If any are 0, the element is definitely not in the set.
21. **Explain the concept of a "hash pointer" and give an example of where it is used.**
- **Answer:** A hash pointer is a pointer to data along with a hash of the data, ensuring that the data can be retrieved and its integrity verified. It is used in blockchain to link blocks of transactions securely.
22. **What are the advantages of using a Merkle Tree for data verification?**
- **Answer:** Merkle Trees allow efficient and secure verification of large datasets by using hashes to represent data blocks, ensuring that the entire dataset can be validated quickly with minimal resources.
23. **What is the primary use of public-key encryption in digital signatures?**
- **Answer:** Public-key encryption is used to encrypt the signature with a private key, ensuring that the signature can be verified by others using the corresponding public key.
24. **How does a Bloom filter handle the situation where a set grows larger?**
- **Answer:** As the set grows larger, the probability of false positives increases because more elements are hashed into the same positions in the bit array. However, Bloom filters are still efficient in terms of memory usage.
25. **What is the impact of using multiple hash functions in a Bloom filter?**
- **Answer:** Using multiple hash functions reduces the chance of false positives by spreading the elements more evenly across the bit array, but it also increases the complexity of the filter.
26. **Explain how the "pigeonhole principle" relates to collision resistance in cryptographic hash functions.**
- **Answer:** The pigeonhole principle states that if the input space is larger than the output space, there must be at least two distinct inputs that map to the same output. Collision resistance makes it difficult to find these collisions in cryptographic hash functions.
27. **How does a Merkle Tree enable "proof of membership"?**
- **Answer:** A Merkle Tree allows a user to prove that a particular leaf node (data) is part of the tree by providing the necessary hashes from the leaf to the root, enabling verification in logarithmic time.
28. **What is a "false positive" in the context of Bloom filters, and why is it problematic?**
- **Answer:** A false positive occurs when the Bloom filter incorrectly reports that an element is part of the set when it is not. This is problematic because it may cause incorrect actions based on erroneous membership tests.
29. **Why is AES a widely used symmetric encryption algorithm?**
- **Answer:** AES is widely used because it provides a high level of security with relatively small key sizes (128, 192, or 256 bits) and is efficient to compute, making it suitable for modern cryptographic applications.

30. **What is the purpose of the private key in the context of digital signatures?**

- **Answer:** The private key is used to create a digital signature by encrypting a hash of the message. This signature can then be verified by anyone with the corresponding public key.

Lecture-4

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. In a distributed system, each node is independent and can operate autonomously.
 - True
 2. A node in a distributed system can only communicate with other nodes that are on the same physical machine.
 - False
Corrected version: A node in a distributed system can communicate with other nodes across the network, regardless of physical machine location.
 3. A distributed system can be considered as a set of processes running on multiple nodes.
 - True
 4. In a distributed system, the network type can either be peer-to-peer (P2P) or client-server.
 - True
 5. In a distributed system, a failure in one node can affect the entire system.
 - False
Corrected version: In a well-designed distributed system, a failure in one node does not necessarily affect the entire system, as it is designed to be fault-tolerant.
-

2. Fill in the blanks: Complete the sentences.

6. A distributed system is made up of multiple _____ that are connected by a communication network.
 - **Answer:** nodes.

7. Each node in a distributed system has its own _____ and non-volatile storage.
 - **Answer:** processor.
 8. In a distributed system, each node communicates using a _____ interface to connect to the network.
 - **Answer:** network.
 9. Concurrent processes in a distributed system can either be _____ or _____ depending on how they share resources.
 - **Answer:** independent, cooperating.
 10. In the context of distributed systems, a process is an execution of a _____ program.
 - **Answer:** sequential.
-

3. Multiple Choice: Choose the correct answer.

11. Which of the following is NOT a characteristic of a distributed system?
 - a) A single control node
 - b) Multiple independent nodes
 - c) Distributed communication
 - **Answer:** a) A single control node
12. What does it mean if a process in a distributed system is considered "cooperating"?
 - a) The process does not share resources with others
 - b) The process exchanges information with other processes
 - c) The process runs on the same machine
 - **Answer:** b) The process exchanges information with other processes
13. Which type of network topology is illustrated by the client-server model?
 - a) P2P
 - b) Centralized
 - c) Hierarchical
 - **Answer:** b) Centralized
14. In a P2P network, nodes are considered to be:
 - a) Clients only
 - b) Servers only
 - c) Equal and autonomous
 - **Answer:** c) Equal and autonomous
15. Which of the following is a fault model commonly considered in distributed systems?
 - a) Reboot failure
 - b) Crash failure
 - c) Power failure
 - **Answer:** b) Crash failure

16. What is a key characteristic of the Byzantine failure model?

- a) Nodes fail to respond
 - b) Nodes send conflicting information to different parts of the system
 - c) Nodes operate in isolation
 - **Answer:** b) Nodes send conflicting information to different parts of the system
-

4. Short Answer Questions

17. What are the main components of a distributed system?

- **Answer:** Nodes (processes), communication network, software, and storage.

18. How does a node in a distributed system communicate with other nodes?

- **Answer:** A node communicates with other nodes using a network interface (NIC) that connects to the communication network.

19. Explain the difference between independent and cooperating processes in a distributed system.

- **Answer:** Independent processes do not share resources or exchange information, while cooperating processes share resources and exchange information to achieve a common goal.

20. What is a "crash failure" in the context of a distributed system?

- **Answer:** A crash failure occurs when a node stops functioning, either due to a hardware or software failure, and becomes unresponsive until further actions are taken.

21. What is the "Byzantine Generals Problem"?

- **Answer:** The Byzantine Generals Problem describes a situation in which a group of generals must agree on a common plan of action, but some of them may act maliciously or send conflicting information to sabotage the decision-making process.

22. What conditions must be met in the Byzantine fault model to ensure a correct decision?

- **Answer:** Loyal generals must decide on the same plan of action, and a small number of traitors should not be able to mislead the loyal generals into making a bad decision.

23. What is the primary difference between crash failure and Byzantine failure in distributed systems?

- **Answer:** Crash failure involves nodes becoming unresponsive without any malicious intent, while Byzantine failure involves nodes sending conflicting or deceptive information, which can be due to malicious actions or software bugs.

24. Why is consensus important in distributed systems?

- **Answer:** Consensus is important because it ensures that all nodes in the system agree on a single shared state or value, which is crucial for maintaining data consistency and ensuring the reliability of the system.

25. How does database replication improve the availability of data in distributed systems?

- **Answer:** Database replication stores data across multiple nodes, ensuring that data is not lost if one or more nodes fail. This increases availability and provides resilience to node failures.
26. **Explain the concept of state machine replication (SMR).**
- **Answer:** State machine replication is a method where a computing machine is replicated across different nodes. Each node processes inputs in the same way, ensuring that all nodes reach the same state, thus maintaining consistency across the system.
27. **What is the role of consensus protocols in state machine replication?**
- **Answer:** Consensus protocols ensure that all nodes in a state machine replication system agree on the same sequence of inputs, which guarantees that all nodes maintain the same state and data.
28. **Why is it necessary for distributed systems to ensure synchronization across nodes?**
- **Answer:** Synchronization ensures that all nodes in a distributed system share the same state or data, preventing inconsistencies and ensuring that the system operates correctly despite node failures or latency.
29. **What happens if a node fails in a distributed system that uses state machine replication?**
- **Answer:** If a node fails, other nodes can continue operating and can be used to recover the state or data of the failed node, ensuring that the system remains operational.
30. **What is the role of atomic broadcast in ensuring consensus in distributed systems?**
- **Answer:** Atomic broadcast ensures that all nodes receive the same set of input messages in the same order, which allows for agreement and consistency in the state of the system, even in the presence of faults.

Lecture – 5: Distributed Systems & Consensus Algorithm

In simple terms, a **consensus protocol** is a set of rules that allows a group of computers (nodes) to agree on a single version of the truth, even if some members are unreliable or trying to cheat.

- Examples of some well-known crash-tolerant consensus protocol are
 - Paxos, Viewstamped Replication, ZooKeeper, and Raft

The shared status is usually a data structure supported by a replicated log

RAFT LEader Selection

-
-
- **The Three Roles:** Every server in the system can only ever be in one of three states: a **Follower**, a **Candidate**, or a **Leader**.
- **Terms (Epochs):** Time in the system is divided into numbered periods called "terms," and the goal is to elect one leader per term.
- **The Waiting Game (Timeouts):** When the servers first start up, they all begin as Followers. Each Follower has a built-in countdown timer, called a timeout (for example, set to 150ms, 200ms, or 300ms).
- **Starting an Election:** If a Follower's timer runs out because it hasn't heard from an active leader, it assumes the leader is gone, changes its role to a **Candidate**, and starts a new election.
- **Asking for Votes:** The new Candidate immediately votes for itself and sends a request to the other servers asking for their votes.
- **Winning the Election:** If the other servers grant their votes and the Candidate receives a **majority** of the votes, it wins and officially becomes the **Leader**. It will remain the leader for the rest of that term. If any other servers were running for leader, they step down and go back to being Followers once they discover the newly elected leader.
- **Handling Ties:** If the votes get split among different Candidates and nobody gets a clear majority, that term ends without a leader. The servers will simply wait for their timers to run out again and start a fresh election for the next term.

RAFT LOG Replication

- **1. The Client Request:** When an outside user (a client) wants to add or update information, they send their request directly to the current **Leader**.
- **2. The Leader Updates:** The Leader takes this new information and appends it to its own log. At this stage, the change is basically penciled in—it is not yet permanent.
- **3. Replicating the Data:** The Leader then sends this new log entry out to all of the **Follower** servers.
- **4. Followers Acknowledge:** Each Follower receives the new information, adds it to their own log, and replies back to the Leader to confirm they got it.
- **5. The Majority Rules (The Commit):** The Leader acts like a tally counter. It waits until it hears back from a **majority** of the servers in the system. Once the majority has replied, the Leader officially "commits" the value to its log, meaning the update is now permanent.
- **6. Finalizing the Sync:** Finally, the Leader sends a message to the Followers letting them know that the entry is official, and the Followers then commit the value into their logs as well.

<https://thesecretlivesofdata.com/raft/>

Primary Rotated + primary unavailable -----> view change

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **Every message broadcast by a valid node will be included in the consensus protocol.**
 - **True**
2. **A consensus protocol is considered safe only when all nodes produce different outputs for the same broadcast.**
 - **False**
Corrected version: A consensus protocol is considered safe only when all nodes produce the same valid output for the same broadcast.
3. **Liveness and availability are interchangeable terms in distributed systems.**
 - **True**
4. **A deterministic consensus protocol can guarantee safety, liveness, and fault tolerance in an asynchronous network.**
 - **False**
Corrected version: A deterministic consensus protocol cannot guarantee all three properties in an asynchronous network (FLP Impossibility).
5. **According to the CAP theorem, a system can achieve consistency and partition tolerance in the presence of network partitions.**
 - **False**
Corrected version: A system cannot achieve both consistency and availability when network partitions occur (CAP theorem).

2. Fill in the blanks: Complete the sentences.

6. The atomic broadcast property of _____ guarantees that messages will be delivered to all valid nodes.
 - Agreement
7. If a consensus protocol guarantees that all nodes produce the same valid output for a broadcast, it is said to ensure _____.
 - Safety
8. The _____ theorem shows that a deterministic consensus protocol cannot guarantee all three properties in an asynchronous network.
 - FLP Impossibility
9. In the CAP theorem, when a network partition occurs, a system must give up _____ or _____.
 - Consistency or Availability
10. Raft's leader election process is crucial for the protocol's _____.
 - Liveness
11. Raft's leader election process ensures that only one _____ is elected at a time in a given term.
 - Leader
12. In the Raft protocol, the leader's role is primarily to _____ and maintain consistency.
 - Accept client requests
13. The Raft protocol's log replication ensures that all nodes' logs remain _____.
 - In sync
14. The fault tolerance condition for Raft is that the number of faulty nodes must be less than _____.
 - $n/2$ (where n is the total number of nodes)
15. A consensus algorithm like PBFT assumes that faulty nodes are controlled by an _____.
 - Adversary

3. Multiple Choice: Choose the correct answer.

16. Which of the following properties ensures that a message will not be broadcast more than once by a valid node?
 - a) Validity
 - b) Agreement
 - c) Integrity
 - d) Total Order

- **Answer:** c) Integrity

17. **What does the FLP Impossibility theorem state?**

- a) A consensus protocol cannot be implemented in a distributed system.
- b) A consensus protocol cannot guarantee fault tolerance.
- c) A deterministic consensus protocol cannot guarantee safety, liveness, and fault tolerance in an asynchronous network.
- d) A consensus protocol cannot guarantee consistency.
- **Answer:** c) A deterministic consensus protocol cannot guarantee safety, liveness, and fault tolerance in an asynchronous network.

18. **Which consensus algorithm is known for its crash-tolerant properties?**

- a) Raft
- b) PBFT
- c) Paxos
- d) All of the above
- **Answer:** d) All of the above

19. **In Raft, the leader's role is to:**

- a) Handle all communication between nodes.
- b) Manage data replication.
- c) Accept client requests.
- d) Both b and c.
- **Answer:** d) Both b and c.

20. **Which consensus algorithm is designed to handle Byzantine nodes?**

- a) Raft
- b) Paxos
- c) PBFT
- d) ZooKeeper
- **Answer:** c) PBFT

21. **In Raft, the term number increases:**

- a) Randomly
- b) When a leader fails
- c) Monotonically
- d) After every consensus round
- **Answer:** c) Monotonically

22. **In PBFT, which of the following roles is responsible for verifying and processing requests?**

- a) Primary
- b) Client
- c) Replica
- d) Follower
- **Answer:** c) Replica

4. Short Answer Questions

23. **What are the four main properties of atomic broadcast?**

- **Answer:**
- Validity
- Agreement
- Integrity
- Total Order

24. **Why is the FLP Impossibility theorem important for consensus protocols?**

- **Answer:** The FLP Impossibility theorem proves that in asynchronous networks, it is impossible for a deterministic consensus protocol to satisfy all three properties (safety, liveness, and fault tolerance) simultaneously, highlighting the trade-offs that must be made.

25. **What is the significance of leader election in the Raft protocol?**

- **Answer:** Leader election ensures that there is a single leader responsible for coordinating the activities of the system. This helps maintain the consistency of the log and ensures the system can continue operating even in the face of leader failures.

26. **What role does the log replication mechanism play in Raft?**

- **Answer:** Log replication ensures that all servers in the Raft cluster have the same log entries, thus maintaining consistency across the distributed system. The leader replicates the logs to followers, and the followers commit to the logs once they receive the necessary number of confirmations.

27. **In Raft, what happens if the majority of nodes do not respond during log replication?**

- **Answer:** The leader cannot commit the log entry, and the process is halted until the majority of nodes confirm receipt of the log entry.

28. **What are the main conditions required for Practical Byzantine Fault Tolerance (PBFT) to function correctly?**

- **Answer:** PBFT requires that the number of Byzantine nodes (f) is less than $n/3$, where n is the total number of participating nodes. This ensures that a quorum can still be formed with honest nodes, even in the presence of faulty nodes.

29. **Explain the three-phase process of PBFT.**

- **Answer:**
- Pre-prepare: The primary node sends a pre-prepare message to replicas.
- Prepare: Replicas validate the message and send a prepare message to other nodes.
- Commit: After receiving enough prepare messages, replicas send a commit message, and the transaction is processed.

30. **What happens during a view change in PBFT?**

- **Answer:** During a view change, if the current primary is unavailable or faulty, replicas can propose a new primary to continue the consensus process.

Lecture – 6: Money and Cryptocurrency

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **Cryptocurrency derives trust from cryptographic primitives, not from chemical or physical properties.**
 - True
 2. **Fiat money relies on cryptographic algorithms to establish its trust.**
 - False
Corrected version: Fiat money derives trust from legal tender statutes.
 3. **In a private blockchain, everyone can write and read data without restrictions.**
 - False
Corrected version: In a private blockchain, read and write access is controlled and restricted to authorized parties.
 4. **Bitcoin uses a central authority for issuing and verifying transactions.**
 - False
Corrected version: Bitcoin is decentralized and does not have a central authority.
 5. **Blockchain can be considered a solution to the problem of trust in centralized systems.**
 - True
-

2. Fill in the blanks: Complete the sentences.

6. **Cryptocurrencies are based on _____, ensuring that trust is derived from mathematical properties.**
 - Answer: cryptography.
7. **Fiat money is backed by _____, while cryptocurrency does not rely on such backing.**
 - Answer: legal tender statutes.
8. **Bitcoin is considered decentralized, distributed, and _____.**
 - Answer: voluntary.
9. **A blockchain is a _____ record of transactions, structured in a specific way.**
 - Answer: distributed.
10. **A _____ blockchain allows everyone to read data but restricts write access.**
 - Answer: hybrid.

3. Multiple Choice: Choose the correct answer.

11. Which of the following is a challenge in creating a virtual coin or note for Bitcoin?
- a) Ensuring there is enough trust in third parties
 - b) Preventing inflation and double-spending
 - c) Decentralizing the network
 - **Answer:** b) Preventing inflation and double-spending
12. Which of the following is a characteristic of Bitcoin?
- a) Centralized issuing authority
 - b) Immune to inflation and bank defaults
 - c) Easy to trace users' transactions
 - **Answer:** b) Immune to inflation and bank defaults
13. What type of blockchain is controlled by a single authority and used between trusted parties?
- a) Public blockchain
 - b) Private blockchain
 - c) Hybrid blockchain
 - **Answer:** b) Private blockchain
14. Which cryptocurrency blockchain is an example of a public blockchain?
- a) Hyperledger
 - b) Corda
 - c) Bitcoin
 - **Answer:** c) Bitcoin
15. What is the purpose of blockchain's consensus mechanism?
- a) To provide a centralized authority
 - b) To ensure all participants trust the same records
 - c) To ensure no one can read the data
 - **Answer:** b) To ensure all participants trust the same records

4. Short Answer Questions

16. Explain the difference between fiat money and cryptocurrency.
- **Answer:** Fiat money is backed by government regulations and legal tender statutes, whereas cryptocurrency is based on cryptographic algorithms and decentralized trust without relying on government or central authority.
17. What are the main challenges of Bitcoin mentioned in the lecture?
- **Answer:**
 - Creation of a virtual coin/note
 - Preventing inflation
 - Validation and legitimacy of coins

- Preventing double-spending
 - Relying on proof instead of trust
18. **How does a blockchain provide a solution to Bitcoin's trust issues?**
- **Answer:** Blockchain provides a decentralized ledger where transactions are verified by multiple participants, eliminating the need for trust in a central authority and ensuring transparency and security.
19. **What is the role of cryptography in cryptocurrency?**
- **Answer:** Cryptography is used to secure transactions, create digital signatures, and verify ownership, ensuring the integrity and trustworthiness of cryptocurrency systems.
20. **How does a public blockchain differ from a private blockchain?**
- **Answer:** A public blockchain is open to everyone, where anyone can join, read, and write data, whereas a private blockchain restricts access and is controlled by a single entity or trusted parties.
21. **What are hybrid blockchains?**
- **Answer:** Hybrid blockchains combine elements of both public and private blockchains, where everyone can read data, but write access is restricted to authorized entities.
22. **What are the main types of blockchain based on read and write access?**
- **Answer:**
 - Public (permissionless) blockchain
 - Private (permissioned) blockchain
 - Hybrid blockchain
 - Consortium blockchain
23. **Why is Bitcoin immune to sovereign censorship?**
- **Answer:** Bitcoin operates in a decentralized network where there is no central authority that can control or censor transactions, making it resistant to governmental interference.
24. **How does a private blockchain maintain security?**
- **Answer:** A private blockchain controls access to the network, sets its own rules and regulations, and restricts read/write access to trusted parties, ensuring higher security and privacy.
25. **What happens to the write access in a hybrid blockchain?**
- **Answer:** Write access is restricted and controlled by a central authority, although everyone can read the data in the blockchain.
26. **What are some examples of cryptocurrencies that use public blockchains?**
- **Answer:** Bitcoin, Ethereum, Solana, Cardano
27. **How do private blockchains benefit organizations?**
- **Answer:** Private blockchains allow organizations to have control over the network, set their own rules, and maintain security while restricting access to trusted participants only.
28. **What is the main advantage of using a hybrid blockchain?**

- **Answer:** A hybrid blockchain allows for controlled write access while providing transparency to all participants, making it suitable for organizations that need some level of privacy but also want public visibility.
29. **What is the concept of a consortium blockchain?**
- **Answer:** A consortium blockchain is a private blockchain controlled by a group of entities, such as a consortium of banks, that collectively manage the network and regulate access.
30. **Summarize the summary table for blockchain types based on read and write access.**
- **Answer:**
 - **Public blockchain:** All can read and write
 - **Private blockchain:** Restricted read and write access
 - **Hybrid blockchain:** All can read, restricted write
 - **Consortium blockchain:** Restricted read and write access

Lecture – 7: Bitcoin - 1

Bitcoin represent users anonymously/pseudonymously

Bitcoin uses an elliptic curve (secp256k1) for its public key cryptography

(radix means the number of unique digits)

. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **Bitcoin is completely decentralized, and no central authority controls its issuance.**
 - **True**
2. **Public keys are used to store private information in the Bitcoin network.**
 - **False**
Corrected version: Public keys are used to create identities for users in the Bitcoin network.
3. **In Bitcoin, users are represented pseudonymously through addresses.**

- True
 - 4. **Bitcoin uses an elliptic curve (secp256k1) for its public key cryptography.**
 - True
 - 5. **A Bitcoin wallet only stores public keys.**
 - False

Corrected version: A Bitcoin wallet stores both private and public keys.
-

2. Fill in the blanks: Complete the sentences.

- 6. **A Bitcoin wallet is a collection of _____.**
 - **Answer:** private keys.
 - 7. **Bitcoin uses the _____ curve for its public key cryptography.**
 - **Answer:** secp256k1.
 - 8. **A Bitcoin address is derived from a _____ using a cryptographic algorithm.**
 - **Answer:** public key.
 - 9. **The _____ function is used to convert a public key to a Bitcoin address.**
 - **Answer:** Base58Check encoding.
 - 10. **Base58 encoding is similar to Base64, but it omits _____.**
 - **Answer:** characters like 0 (zero), O (capital O), l (lowercase L), I (capital I), '+', and '/.
-

3. Multiple Choice: Choose the correct answer.

- 11. **What is the main advantage of using a hardware wallet over a software wallet?**
 - a) It is easier to use
 - b) It stores private keys in a secure hardware device
 - c) It can be used offline only
 - **Answer:** b) It stores private keys in a secure hardware device
- 12. **Which of the following is a key feature of cold wallets?**
 - a) They are always connected to the internet
 - b) They are used for daily trading
 - c) They store private keys offline for extra security
 - **Answer:** c) They store private keys offline for extra security
- 13. **What is the primary purpose of a mnemonic phrase in an HD wallet?**
 - a) To create a secure password
 - b) To generate a seed that can regenerate the private keys
 - c) To encode the public key
 - **Answer:** b) To generate a seed that can regenerate the private keys
- 14. **What happens if the private key associated with a Bitcoin address is lost?**
 - a) The Bitcoins are returned to the owner

- b) The Bitcoins are permanently lost
 - c) The Bitcoins are accessible by anyone who has the public key
 - **Answer:** b) The Bitcoins are permanently lost
15. **Which type of wallet stores private keys on an external device, making it more secure against malware attacks?**
- a) Hot wallet
 - b) Paper wallet
 - c) Hardware wallet
 - **Answer:** c) Hardware wallet
16. **What is the main disadvantage of a brain wallet?**
- a) It is difficult to memorize the passphrase
 - b) It is prone to malware attacks
 - c) It requires an internet connection to work
 - **Answer:** a) It is difficult to memorize the passphrase
-

4. Short Answer Questions

17. **How does Bitcoin represent users in the network?**
- **Answer:** Bitcoin represents users pseudonymously using addresses generated through public key cryptography.
18. **Explain the process of generating a Bitcoin address from a public key.**
- **Answer:** A Bitcoin address is generated by applying the SHA256 and RIPEMD160 hashing algorithms to the public key. Then, the address is encoded using Base58Check encoding.
19. **What are the main types of Bitcoin wallets discussed in the lecture?**
- **Answer:**
 - Hot wallet
 - Paper wallet (cold wallet)
 - Hardware wallet
 - Brain wallet
20. **What is the role of the Certificate Authority (CA) in a private blockchain system?**
- **Answer:** In a private blockchain, the Certificate Authority (CA) vets the identities of users, ensuring they are trusted and authenticated before they can participate in the network.
21. **What are the main components of a Bitcoin wallet?**
- **Answer:** A Bitcoin wallet contains a private key and a public key. The private key is used for signing transactions, and the public key is used to receive funds.
22. **What is the significance of a mnemonic phrase in an HD wallet?**
- **Answer:** A mnemonic phrase, typically consisting of 12 words, serves as a seed to regenerate all the keys in an HD wallet, enabling users to back up and restore their wallet.
23. **Why are private keys kept encrypted in a Bitcoin wallet?**

- **Answer:** Private keys are encrypted to ensure their security, preventing unauthorized access and theft, especially in hot wallets that are connected to the internet.
24. **What is the difference between a hot wallet and a cold wallet?**
- **Answer:** A hot wallet is always connected to the internet, making it more convenient but less secure, whereas a cold wallet is offline and provides more security for long-term storage.
25. **Explain the process of creating a private key using a passphrase in a brain wallet.**
- **Answer:** A passphrase (seed phrase) is hashed using a cryptographic algorithm to generate the private key. This private key is then used to derive the public key and Bitcoin address.
26. **What is the advantage of using hardware wallets over software wallets?**
- **Answer:** Hardware wallets store private keys in a secure, offline device, making them immune to computer viruses and malware that can compromise software wallets.
27. **What happens when a brain wallet is forgotten?**
- **Answer:** If a brain wallet is forgotten and the seed phrase is not recorded, the Bitcoins associated with it are permanently lost.
28. **How does Base58 encoding improve upon Base64 encoding in Bitcoin addresses?**
- **Answer:** Base58 encoding removes visually similar characters (e.g., 0, O, l, I) and characters that could be misinterpreted in certain fonts, making it more user-friendly for Bitcoin addresses.
29. **What is the role of the "checksum" in the Base58Check encoding process for Bitcoin addresses?**
- **Answer:** The checksum ensures the integrity of the Bitcoin address by providing a way to detect errors in the address, which is added to the address before encoding in Base58.
30. **Why is Base58 encoding used instead of Base64 for Bitcoin addresses?**
- **Answer:** Base58 is used because it avoids characters that can be confusing or easily misread, improving usability, especially when addresses are manually written or scanned.

Lecture – 8: Bitcoin - 2

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **Bitcoin nodes communicate in a centralized fashion with one superior node controlling the network.**

- **False**
Corrected version: Bitcoin nodes communicate in a decentralized fashion, meaning all nodes are equal.
 - 2. **In Bitcoin, new nodes can join the network at any time.**
 - **True**
 - 3. **A node discovery process is also known as bootstrapping.**
 - **True**
 - 4. **In Bitcoin, the version message helps nodes check the compatibility of the Bitcoin network.**
 - **True**
 - 5. **The Bitcoin P2P network uses a fixed topology with only one central node.**
 - **False**
Corrected version: Bitcoin P2P network uses a random topology.
-

2. Fill in the blanks: Complete the sentences.

- 6. **The Bitcoin network runs on TCP port _____.**
 - **Answer:** 8333.
 - 7. **The first message exchanged between two Bitcoin peers is the _____ message.**
 - **Answer:** version.
 - 8. **A Bitcoin transaction has a list of _____ and _____.**
 - **Answer:** inputs, outputs.
 - 9. **The Bitcoin P2P network uses a _____ protocol for node discovery and propagation of messages.**
 - **Answer:** gossip.
 - 10. **If a node is not responsive for _____ hours, it is removed from the Bitcoin network.**
 - **Answer:** 3.
-

3. Multiple Choice: Choose the correct answer.

- 11. **Which of the following is the primary purpose of the "version" message in Bitcoin?**
 - a) To verify transactions
 - b) To establish node compatibility
 - c) To generate new addresses
 - **Answer:** b) To establish node compatibility
- 12. **Which of the following methods is used for bootstrapping a new node in the Bitcoin network?**
 - a) Manual entry of node IPs

- b) Hard-coded DNS servers
 - c) Random IP generation
 - **Answer:** b) Hard-coded DNS servers
13. **What does the "verack" message signify in the node discovery process?**
- a) A new block has been added to the blockchain
 - b) The node is compatible and acknowledges the connection
 - c) The transaction has been validated
 - **Answer:** b) The node is compatible and acknowledges the connection
14. **What is a major advantage of Bitcoin's decentralized P2P network?**
- a) Faster transaction validation
 - b) No single point of failure
 - c) Centralized control over network protocols
 - **Answer:** b) No single point of failure
15. **Which of the following is a method used for connecting Bitcoin nodes?**
- a) Using a centralized server
 - b) DNS service for node IP addresses
 - c) Through encrypted VPN
 - **Answer:** b) DNS service for node IP addresses
16. **What is the first message that a new Bitcoin node sends to establish communication with another peer?**
- a) Transaction message
 - b) Version message
 - c) Blockchain request
 - **Answer:** b) Version message
-

4. Short Answer Questions

17. **What are the key components of the Bitcoin P2P network?**
- **Answer:** Nodes, blockchain, consensus protocol, and communication rules such as gossip and node discovery.
18. **Explain the process of bootstrapping a new node into the Bitcoin network.**
- **Answer:** A new node discovers other nodes in the network using DNS servers, seed addresses, or manually provided addresses. It then exchanges version and verack messages to establish compatibility with other peers.
19. **What is the role of DNS servers in the Bitcoin P2P network?**
- **Answer:** DNS servers provide a static list of IP addresses for stable Bitcoin nodes, helping new nodes discover peers to connect with.
20. **What is the purpose of the version message in the Bitcoin node discovery process?**
- **Answer:** The version message is sent by a node to check if another node is compatible with the Bitcoin network protocol.
21. **How does Bitcoin ensure that all nodes are equal in the network?**

- **Answer:** Bitcoin uses a decentralized protocol where no single node is superior, and all nodes communicate on an equal footing in a peer-to-peer fashion.
22. **What happens if a node fails to respond for more than 3 hours in the Bitcoin network?**
- **Answer:** The non-responsive node is removed from the Bitcoin network.
23. **How does the Bitcoin network handle transactions and blocks?**
- **Answer:** Transactions and blocks are validated and propagated across the network using a decentralized communication protocol. Every node in the network ensures consistency and verifies transactions before adding them to the blockchain.
24. **Describe the significance of "lock_time" in Bitcoin transactions.**
- **Answer:** "Lock_time" defines the time or block height at which a transaction can be added to the blockchain. A value of 0 means the transaction can be added immediately.
25. **What is the primary challenge that Bitcoin addresses through its decentralized network?**
- **Answer:** Bitcoin addresses the issue of central authority control and the single point of failure in financial systems by enabling secure, decentralized transactions.
26. **What is the difference between a "spent" and "unspent" transaction output (STXO and UTXO)?**
- **Answer:** An unspent transaction output (UTXO) refers to outputs that have not been used in any other transaction, while a spent transaction output (STXO) refers to outputs that have already been consumed in a subsequent transaction.
27. **Explain the concept of "CoinJoin" in Bitcoin transactions.**
- **Answer:** CoinJoin is a method for combining multiple Bitcoin transactions into a single transaction to enhance privacy by obfuscating the transaction paths.
28. **What is the role of "scriptSig" and "scriptPubKey" in Bitcoin transactions?**
- **Answer:** "scriptSig" is the input script that provides the signature to redeem an output, while "scriptPubKey" is the output script that specifies the conditions to redeem the coins.
29. **Why does Bitcoin use a transaction-based ledger instead of an account-based ledger?**
- **Answer:** Bitcoin uses a transaction-based ledger to simplify the tracking of transactions and avoid the complexities of maintaining account balances across a distributed network. This approach ensures that users can define conditional transactions using Bitcoin Script.
30. **How does a wallet maintain and calculate a user's Bitcoin balance?**
- **Answer:** A wallet scans the blockchain and aggregates the value of any unspent transaction outputs (UTXOs) that the wallet can spend with the keys it controls.

Lecture – 9: Bitcoin - 3

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **Bitcoin transactions are identified by a unique transaction ID (TxID).**
 - **True**
 2. **The transaction fee is explicitly defined as a separate field in the Bitcoin transaction.**
 - **False**
Corrected version: The transaction fee is implicitly defined as the difference between the sum of inputs and the sum of outputs.
 3. **A transaction input references a previous unspent transaction output (UTXO) to be spent in the transaction.**
 - **True**
 4. **A transaction output in Bitcoin contains a locking script (scriptSig) and the amount of coins being transferred.**
 - **False**
Corrected version: A transaction output contains a locking script (scriptPubKey) and the amount of coins being transferred.
 5. **ScriptSig is used to lock the value of a transaction output.**
 - **False**
Corrected version: scriptSig is used to unlock the value of a transaction input, not output.
-

2. Fill in the blanks: Complete the sentences.

6. **A Bitcoin transaction contains _____ and _____ to define how coins are transferred.**
 - **Answer:** inputs, outputs.
7. **Bitcoin transaction inputs reference _____ and use _____ to unlock them.**
 - **Answer:** unspent transaction outputs (UTXOs), scriptSig.
8. **The Bitcoin blockchain uses _____ to verify the ownership of a Bitcoin address.**
 - **Answer:** digital signatures.
9. **The sum of inputs minus the sum of outputs determines the _____ of a Bitcoin transaction.**
 - **Answer:** fee.
10. **The TxID in Bitcoin is created by double hashing the _____ and _____ of the transaction.**
 - **Answer:** inputs, outputs.

3. Multiple Choice: Choose the correct answer.

11. What does the "scriptPubKey" do in a Bitcoin transaction?
- a) It provides a signature for the transaction
 - b) It locks the transaction output to a specific Bitcoin address
 - c) It verifies the transaction
 - **Answer:** b) It locks the transaction output to a specific Bitcoin address
12. What is the main function of the scriptSig in Bitcoin transactions?
- a) To unlock a transaction input
 - b) To lock a transaction output
 - c) To verify the ownership of the Bitcoin address
 - **Answer:** a) To unlock a transaction input
13. Which of the following does NOT require a scriptSig to be validated in a Bitcoin transaction?
- a) Coinbase transaction
 - b) Regular transactions
 - c) Block header
 - **Answer:** a) Coinbase transaction
14. What is the main use of the "lock_time" field in Bitcoin transactions?
- a) To set the transaction fee
 - b) To set when the transaction is valid
 - c) To limit the time the transaction is valid for mining
 - **Answer:** b) To set when the transaction is valid
15. In Bitcoin transactions, which type of output is considered an unspent transaction output (UTXO)?
- a) Outputs that have not been consumed by other transactions
 - b) Outputs that have been used in a new transaction
 - c) Outputs that are part of the block header
 - **Answer:** a) Outputs that have not been consumed by other transactions
16. How is the fee for a Bitcoin transaction calculated?
- a) Fee = Sum of outputs – sum of inputs
 - b) Fee = Sum of inputs – sum of outputs
 - c) Fee = Total value of inputs
 - **Answer:** b) Fee = Sum of inputs – sum of outputs
-

4. Short Answer Questions

17. What are the three main elements contained in a Bitcoin transaction input?
- **Answer:**
 - 1. Transaction ID (TxID) referencing the previous transaction

2. Output index (vout) identifying which UTXO from the transaction is being spent
3. ScriptSig, which satisfies the conditions placed on the UTXO to unlock it
4. **Explain how the Bitcoin transaction ID (TxID) is created.**
 - **Answer:** TxID is created by double hashing the serialized inputs, outputs, and other data of the transaction using the SHA256 hashing algorithm.
19. **How does Bitcoin ensure that only the rightful owner can spend the Bitcoin in a transaction output?**
 - **Answer:** By using a locking script (scriptPubKey) that defines the conditions to redeem the Bitcoin. The rightful owner proves ownership by providing the correct unlocking script (scriptSig) containing their digital signature.
20. **Describe the difference between UTXO (Unspent Transaction Output) and STXO (Spent Transaction Output).**
 - **Answer:** UTXOs are outputs that have not been consumed or spent in a subsequent transaction, while STXOs are outputs that have already been used as inputs in another transaction.
21. **What is the role of the Merkle root in Bitcoin blocks?**
 - **Answer:** The Merkle root is a hash that summarizes all the transactions in a block and is included in the block header. It allows for efficient and secure verification of the integrity of all the transactions in the block.
22. **How does the wallet calculate the balance of a Bitcoin address?**
 - **Answer:** The wallet scans the blockchain for all unspent transaction outputs (UTXOs) that it controls, aggregates their values, and calculates the total balance of the address.
23. **What is the importance of the "n" field in Bitcoin transactions?**
 - **Answer:** The "n" field refers to the index of the output in the previous transaction and helps to identify which output is being consumed in the current transaction.
24. **How does Bitcoin handle "change" in a transaction?**
 - **Answer:** If a user wants to send less than the total amount of the UTXO being used, Bitcoin creates a new output for the "change" and sends it back to the user's wallet.
25. **Why is a scriptSig used in Bitcoin transactions?**
 - **Answer:** scriptSig is used to provide proof of ownership of the UTXO by including the digital signature and other data to unlock the transaction input and authorize the transfer of funds.
26. **Explain the concept of "coinbase" transaction.**
 - **Answer:** A coinbase transaction is the first transaction in a block that creates new Bitcoins. It does not have any inputs but has one or more outputs, typically giving the miner the block reward.
27. **What role does the "nonce" play in Bitcoin transactions and blocks?**
 - **Answer:** The nonce is used in the Proof-of-Work algorithm to help miners find a valid block hash that meets the network's difficulty target.
28. **How are multiple transactions combined into a block in Bitcoin?**
 - **Answer:** Multiple transactions are grouped together in a block, each with its own TxID. These transactions, along with the block header, are hashed and added to the blockchain.
29. **What is the "difficulty target" field in the Bitcoin block header?**

- **Answer:** The difficulty target defines the level of difficulty required for miners to find a valid hash for the block. It adjusts approximately every two weeks to maintain a consistent block time of 10 minutes.
30. **What happens when a Bitcoin transaction is confirmed?**
- **Answer:** A transaction is confirmed once it is included in a valid block that has been added to the blockchain. After confirmation, the transaction is considered final and irreversible.

Lecture – 10: Bitcoin - 4

1. True or False: Determine if the following sentences are true or false. For any false statement, write its correct form.

1. **In Bitcoin, miners receive rewards for solving the cryptographic puzzle and creating new blocks.**
 - True
 2. **Proof of Work (PoW) is a cryptographic puzzle that miners must solve to add a new block to the Bitcoin blockchain.**
 - True
 3. **The difficulty of the Bitcoin mining puzzle is fixed and does not change over time.**
 - False
Corrected version: The difficulty of the Bitcoin mining puzzle is adjusted every 2016 blocks to maintain a constant block time of 10 minutes.
 4. **The "nonce" in a Bitcoin block is used to store transaction information.**
 - False
Corrected version: The "nonce" in a Bitcoin block is used to solve the Proof of Work puzzle by finding a value that makes the block hash less than the difficulty target.
 5. **Miners in Bitcoin earn rewards only through transaction fees, not block rewards.**
 - False
Corrected version: Miners earn rewards both through transaction fees and the block reward (newly minted Bitcoins).
-

2. Fill in the blanks: Complete the sentences.

6. Bitcoin mining requires solving a cryptographic puzzle using the _____ algorithm.
 - Answer: SHA256.
 7. The purpose of the _____ is to maintain the average time between blocks at 10 minutes.
 - Answer: difficulty adjustment.
 8. The first block in the Bitcoin blockchain is called the _____ block.
 - Answer: genesis.
 9. The Bitcoin block reward is halved every _____ blocks, approximately every 4 years.
 - Answer: 210,000.
 10. The Bitcoin network adjusts the difficulty every _____ blocks to ensure a constant block time.
 - Answer: 2016.
-

3. Multiple Choice: Choose the correct answer.

11. Which of the following is used to prevent Bitcoin block rewards from increasing exponentially?
 - a) Difficulty adjustment
 - b) SHA256 algorithm
 - c) Proof of Stake
 - Answer: a) Difficulty adjustment
12. What does a miner need to find in order to create a valid Bitcoin block?
 - a) A valid public key
 - b) A hash less than the difficulty target
 - c) A new address
 - Answer: b) A hash less than the difficulty target
13. How often does the Bitcoin block reward halve?
 - a) Every 100,000 blocks
 - b) Every 210,000 blocks
 - c) Every 500,000 blocks
 - Answer: b) Every 210,000 blocks
14. Which component of a Bitcoin block is used to link blocks together in a chain?
 - a) Nonce
 - b) Merkle root
 - c) Previous block hash
 - Answer: c) Previous block hash
15. What happens when two miners find valid blocks at the same time?
 - a) The network chooses the block based on the nonce value

- b) A fork occurs, and the network has two competing chains
 - c) The miners must submit a new solution
 - **Answer:** b) A fork occurs, and the network has two competing chains
16. **What is the main reason for increasing the mining difficulty in Bitcoin?**
- a) To control the supply of Bitcoin
 - b) To ensure blocks are mined at regular intervals
 - c) To reward miners more frequently
 - **Answer:** b) To ensure blocks are mined at regular intervals
-

4. Short Answer Questions

17. **Explain the Proof of Work (PoW) puzzle in Bitcoin mining.**
- **Answer:** Proof of Work (PoW) is a cryptographic puzzle that requires miners to find a hash value for a block that is lower than a specified difficulty target. The process involves repeatedly changing the nonce value until the hash of the block header meets the target. This ensures that mining is computationally expensive and prevents spam attacks.
18. **What happens when the Bitcoin network experiences an increase in mining power?**
- **Answer:** When mining power increases, more blocks are produced in a shorter period of time. To maintain the block time of 10 minutes, the difficulty is adjusted upward after every 2016 blocks, making it harder to mine subsequent blocks.
19. **Why is the Bitcoin mining process described as a "lottery"?**
- **Answer:** Bitcoin mining is often compared to a lottery because miners randomly guess values (nonces) and compete to find a valid block hash that meets the difficulty target. The process requires a lot of computational power and luck, much like trying to win a lottery.
20. **What is the role of the "nonce" in the Bitcoin mining process?**
- **Answer:** The nonce is a number that miners change in the block header to try to find a valid hash that meets the difficulty target. The nonce allows miners to modify the hash calculation and "solve" the Proof of Work puzzle.
21. **What is the relationship between difficulty and the time taken to mine Bitcoin blocks?**
- **Answer:** The difficulty is adjusted every 2016 blocks to ensure that the time between blocks remains around 10 minutes. If blocks are being mined too quickly, difficulty increases; if they are mined too slowly, difficulty decreases.
22. **Why does Bitcoin use a halving mechanism for the block reward?**
- **Answer:** The halving mechanism reduces the rate at which new Bitcoins are created, which gradually decreases the inflation rate of Bitcoin. It ensures that the total supply of Bitcoin will be capped at 21 million, mimicking the scarcity of precious metals like gold.
23. **What is the genesis block in Bitcoin?**

- **Answer:** The genesis block is the first block in the Bitcoin blockchain. It is hardcoded into the Bitcoin software and has no predecessor. It serves as the foundation for all subsequent blocks in the chain.

24. What role does the Merkle root play in a Bitcoin block?

- **Answer:** The Merkle root is a hash that summarizes all the transactions in a block. It is used to efficiently verify the integrity of the transactions in the block and is included in the block header.

25. Explain the concept of a "fork" in the Bitcoin network.

- **Answer:** A fork occurs when two miners produce a valid block at the same time, causing the network to temporarily have two competing versions of the blockchain. Nodes will eventually adopt the longer chain (the one with the most accumulated proof of work) and discard the shorter one.

26. What happens to transactions in an orphaned block?

- **Answer:** Transactions in an orphaned block are returned to the mempool if the block is discarded. They are then available to be included in future blocks.

27. How does the Bitcoin network resolve forks?

- **Answer:** When a fork occurs, the Bitcoin network chooses the chain with the most accumulated proof of work (longest chain) as the valid chain. Transactions from orphaned blocks are returned to the mempool and included in future blocks.

28. How does the difficulty adjustment ensure that Bitcoin's block time remains constant?

- **Answer:** The difficulty is adjusted every 2016 blocks based on the time it took to mine the last 2016 blocks. If blocks are being mined too quickly, the difficulty increases; if too slowly, it decreases, ensuring an average block time of 10 minutes.

29. What is the maximum number of Bitcoins that will ever be mined?

- **Answer:** The maximum number of Bitcoins that will ever be mined is 21 million, which is the total supply limit set by Bitcoin's protocol.

30. Explain the Bitcoin mining "arms race" described in the lecture.

- **Answer:** The "arms race" refers to the competitive nature of mining, where miners continuously upgrade their hardware to increase their hashing power in order to outpace other miners and secure more block rewards. This increases the total network hashing power and adjusts the difficulty, making mining more expensive and energy-intensive.